

Transformixer: Attention-Based Variate Mixing without Positional Bias for Multivariate Forecasting

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Task: multivariate long-term forecasting

lookback $X \in \mathbb{R}^{L \times C} \rightarrow$ horizon $\hat{Y} \in \mathbb{R}^{H \times C}$

Challenge

Models must capture *temporal* dynamics **and** *cross-variate* dependencies.

Especially hard when C is large (e.g., Traffic: $C=862$) and channels have **no natural order**.

Recent mixers (e.g., xLSTM-Mixer) refine a linear temporal forecast with a joint variate mixer.

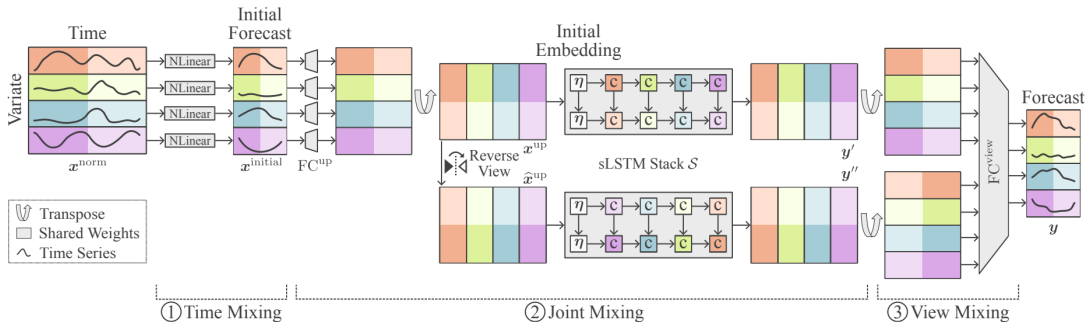
But recurrent mixing along channels is order-sensitive.

- ① ReVlN normalization
- ② Shared **NLinear** temporal forecast (efficient)
- ③ **sLSTM** mixes along the *variate* axis (+ memory tokens)
- ④ Forward + **reverse** multi-view fusion

Inductive-bias issue

Channel indices (sensors, meters, roads) are often an **arbitrary ordering**.
Sequential mixing depends on that order and needs a reverse view to reduce artifacts.

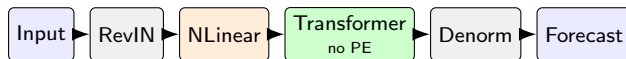
xLSTM-Mixer architecture



Same temporal pipeline (RevIN + NLinear); Transmixer replaces joint/view mixing with a PE-free Transformer.

Keep what works: RevIN + shared NLinear temporal mixing.

Replace the ordered sLSTM mixer with a PE-free Transformer over variate tokens.



- Variates treated as a **set** (permutation-equivariant)
- Direct pairwise interactions via multi-head attention
- No memory tokens, no forward/reverse multi-view

vs. xLSTM-Mixer (what changes)

Component	xLSTM-Mixer	Transformixer
RevIN + NLinear	yes	yes
Variate mixer	sLSTM (+ mem.)	Transformer
Positional bias on variates	via recurrence	none
Forward/reverse views	yes	no
Memory tokens	yes	no

Related but different from iTransformer

iTransformer = full inverted backbone.

Transformixer = attention as a *drop-in mixer block* after NLinear.

Why Transformer mixing? (not only equivariance)

- ① **Direct global interactions** — $O(1)$ path length between any pair of variates (sLSTM propagates sequentially along an arbitrary order)
- ② **Multi-head specialization** — different dependency patterns in parallel
- ③ **Simpler architecture** — drop multi-view + memory tokens
- ④ **Hardware-parallel over C** — we measure training it/s; $O(C^2)$ caveat remains

Hypothesis: PE-free attention should help most when C is large (Traffic); PE is a separate, dataset-dependent knob.

Datasets: ECL ($C=321$), Traffic ($C=862$)

Protocol: $L=H=96$; MSE/MAE;
multivariate

Baselines: RLinear, Informer, PatchTST,
iTransformer, xLSTM-Mixer

Fairness: baselines author-tuned;
Transformixer not independently tuned

Controlled ablation (same recipe)

A: NLinear only B: xLSTM FULL

C: Transformixer (no PE, default)

D: Transformixer (+ PE)

Scope

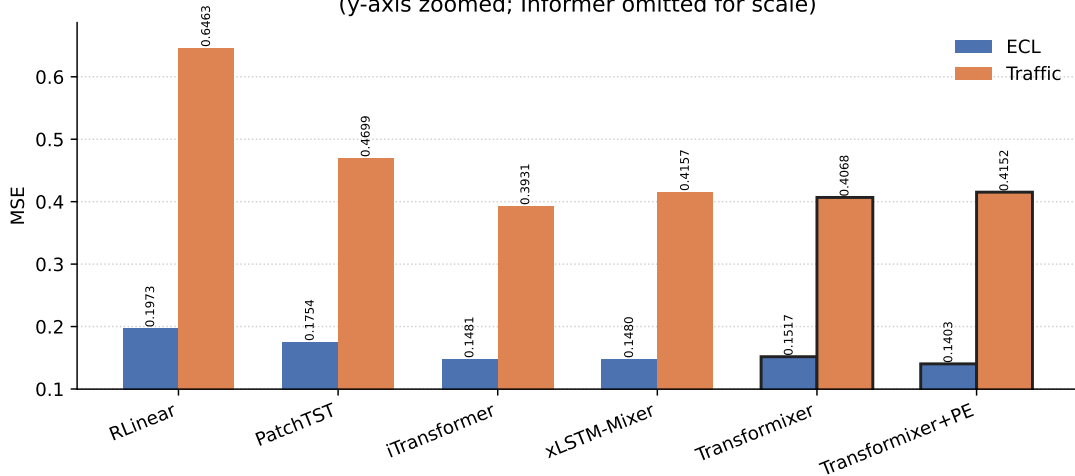
2 datasets, 1 horizon, single runs. Main table includes both Transformixer variants (C and D).

Model	ECL		Traffic	
	MSE	MAE	MSE	MAE
RLinear	0.1973	0.2740	0.6463	0.3844
Informer	0.3263	0.4068	1.0140	0.5624
PatchTST	0.1754	0.2603	0.4699	0.3005
iTransformer	0.1481	0.2398	0.3931	0.2687
xLSTM-Mixer	<u>0.1480</u>	<u>0.2352</u>	0.4157	<u>0.2607</u>
Transformixer	0.1517	0.2364	<u>0.4068</u>	0.2558
Transformixer+PE	0.1403	0.2315	0.4152	0.2639

- **ECL:** Transformixer+PE best overall; PE-free default slightly behind xLSTM / iTransformer
- **Traffic:** PE-free default beats xLSTM-Mixer; **best MAE** overall; +PE slightly worse
- Positional bias is dataset-dependent (also confirmed in controlled ablation)

Results at a glance

Long-term forecasting (MSE), lookback/horizon = 96
(y-axis zoomed; Informer omitted for scale)

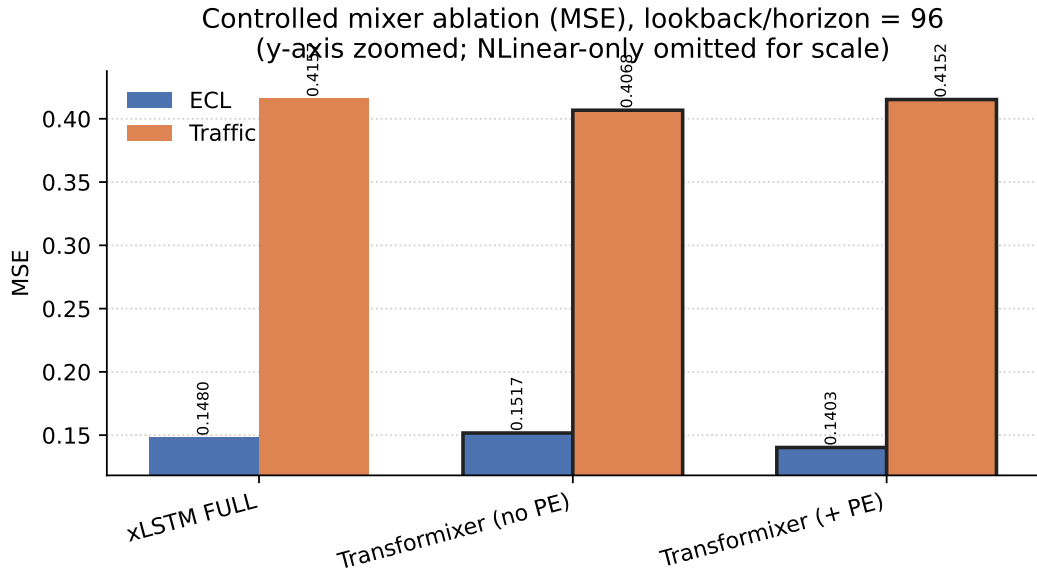


Informer omitted for scale; y-axis zoomed around competitive models.

Mixer ablation (shared RevIN+NLinear)

Variate mixer	ECL		Traffic	
	MSE	MAE	MSE	MAE
A: NLinear only	0.2016	0.2731	0.6713	0.3776
B: xLSTM FULL	<u>0.1480</u>	<u>0.2352</u>	0.4157	<u>0.2607</u>
C: Transformixer (no PE)	0.1517	0.2364	0.4068	0.2558
D: Transformixer (+ PE)	0.1403	0.2315	<u>0.4152</u>	0.2639

- Mixer $\gg$ NLinear-only on both datasets (esp. Traffic)
- **Traffic:** PE-free (C) best; fixed PE (D) hurts
- **ECL:** fixed PE (D) best; index-aware bias can help

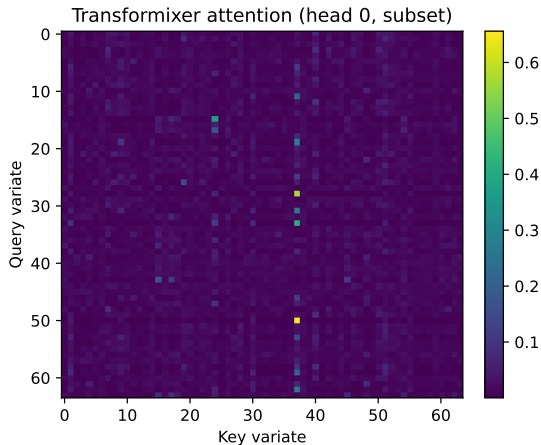


Permutation probe (Traffic checkpoints)

Model	Identity		Over 9 orders (mean $\pm$ std)	
	MSE	MAE	MSE	MAE
xLSTM-Mixer	1.3463	0.9224	1.3486 $\pm$ 0.0015	0.9233 $\pm$ 0.0005
Transformixer	1.2841	0.9019	1.2865 $\pm$ 0.0014	0.9027 $\pm$ 0.0005

- PE-free checkpoint: nearly invariant (equivariance)
- xLSTM mild drift (multi-view helps), but **always worse**
- Relative gap $\approx 4.6\%$ MSE / 2.2% MAE on this probe

Attention visualization



What we see

- Sparse, non-uniform attention
- Hub-like key variates
- Not purely diagonal

Supports *direct pairwise* cross-variate mixing.

Dataset	C	xLSTM-Mixer	Transformixer
ECL	321	3.53 it/s	4.67 it/s
Traffic	862	1.32 it/s	1.33 it/s

- **ECL:** Transformixer $\approx 32\%$ higher training throughput
- **Traffic:** essentially tied — quadratic attention cost catches up at high C
- Progress-bar it/s (fwd+bwd, data loading, logging); **not** FLOPs or inference latency
- No general wall-clock superiority claim

Limitations (stated openly)

- Only **2 datasets**, horizon **96**, **single** runs
- Main baselines use author-tuned hparams; Transformixer not independently tuned
- Throughput is single-run it/s on L4; FLOPs / peak memory / inference not profiled

Still useful

Main results and ablation agree: mixer is necessary; PE vs equivariance is dataset-dependent.
Future work: learnable global permutations (e.g., Gumbel-Sinkhorn).

- ① **Transformixer** = NLinear temporal mix + PE-free Transformer variate mixer (default)
- ② Main results: +PE best on ECL; PE-free best Traffic MAE; bias is dataset-dependent
- ③ Controlled ablation: mixer necessary; same PE vs equivariance pattern
- ④ Throughput on L4: faster on ECL, tied on Traffic (parallelism vs $O(C^2)$)
- ⑤ Future: learnable channel orderings via differentiable relaxations

Thank you

Questions?